

SEMESTER-IV

UMA011: NUMERICAL ANALYSIS				
	L	T	P	Cr
	3	0	2	4.0
Course Objective: The main objective of this course is to motivate the students to understand and learn various numerical techniques to solve mathematical problems representing various engineering, physical and real-life problems.				
Syllabus Floating-Point Numbers: Floating-point representation, rounding, chopping, error analysis, conditioning and stability. Non-Linear Equations: Bisection, secant, fixed-point iteration method and its convergence analysis, Newton method for simple and multiple roots, and order of convergence. Linear Systems and Eigen-Values: Gauss elimination method using pivoting strategies, LU decomposition, Gauss-Seidel and successive-over-relaxation (SOR) methods and their convergence, Rayleigh's power method for Eigen-values and Eigen-vectors. Interpolation and Approximations: Finite differences, Newton's forward and backward interpolation, Lagrange and Newton's divided difference interpolation formulas with error analysis, least square approximations. Numerical Integration: Newton-Cotes quadrature formulae (Trapezoidal and Simpson's rules) and their error analysis, Gauss-Legendre quadrature formulae. Differential Equations: Solution of initial value problems using Taylor series, Euler's and Runge-Kutta methods of order four, system of first-order differential equations.				
Laboratory Work Lab experiments will be set in consonance with materials covered in the theory. Implementation of numerical techniques using MATLAB .				
Course Learning Outcomes (CLOs) The students will be able to: 1. understand the errors, source of error and its effect on any numerical computations. 2. learn how to obtain numerical solution of nonlinear equations using bisection, secant, Newton, and fixed-point iteration methods. 3. solve system of linear equations numerically using direct and iterative methods. 4. understand how to approximate the functions using interpolating polynomials. 5. learn how to solve definite integrals and initial value problems numerically.				
Text Books 1. Applied Numerical Analysis, Gerald F. C. and Wheatley O. P., Pearson 7th ed., 2003. 2. Numerical Methods for Scientific and Engineering Computation, Jain K. M., Iyengar K. R. S. and Jain K. R., New Age International Publishers 6th ed., 2012. 3. Numerical Methods for Engineers, Steven C. Chappra, McGraw-Hill Higher Education, 7th edition, 2014.				

Approved in 109th meeting of the Senate held on March 16, 2023. Revised in 112th and 113th meeting of the Senate held on March 11 and September 7, 2024, respectively. Further, revision Approved in 1st meeting of Academic Council held on September 11, 2025.

Reference Books

1. Numerical Methods for Mathematics, Science and Engineering, Mathew H. J., Prentice Hall, 2nd ed., 1992.
2. Numerical Analysis, Burden L. R. and Faires D. J. Brooks Cole 9th ed., 2011.
3. Elementary Numerical Analysis, Atkinson K. and Han H., John Willey & Sons, 3rd ed., 2004.

Evaluation Scheme:

Sr.No.	Evaluation Elements	Weightage (%)
1.	MST	25
2.	EST	40
3.	Sessionals (May include assignments/quizzes)	15
4	Laboratory evaluation	20

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